Chapter 3

Observed Regularities

Beyond the classification, several quantitative regularities emerged across all trials.

3.1 Internal Color Periodicity

Along any directional line, the height pattern repeats periodically without variation. For example, a line in direction $(2, 1)$ always follows the sequence (Blue, White, Blue) from left to right.

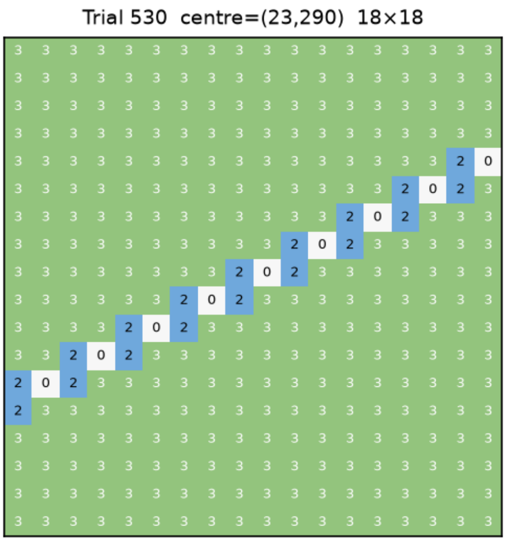

Figure 3.1: Directional pattern $(2, 1)$, showing the periodic sequence (Blue, White, Blue).

3.2 Frequency of Occurrence

Lines with near-horizontal or near-vertical slopes appear less often. Most observed lines are horizontal, vertical, or at 45° (slope ± 1). Lines with slopes $1/2$, $1/3$, 2, or 3 are rare.

3.3 Vector Sum Rule

When several directional lines meet at a point, their direction vectors sum to zero. This vector sum rule governs the pattern geometry.

Starting from the two basic directions $(1, 0)$ and $(0, 1)$, the vector sum rule generates the remaining observed directions $(1, 1)$, $(2, 1)$, $(1, 2)$, $(3, 1)$, $(1, 3)$. The examples below show how each new direction arises from an intersection of two earlier ones.

The intersection of $(1, 0)$ and $(0, 1)$ yields $(1, 1)$.

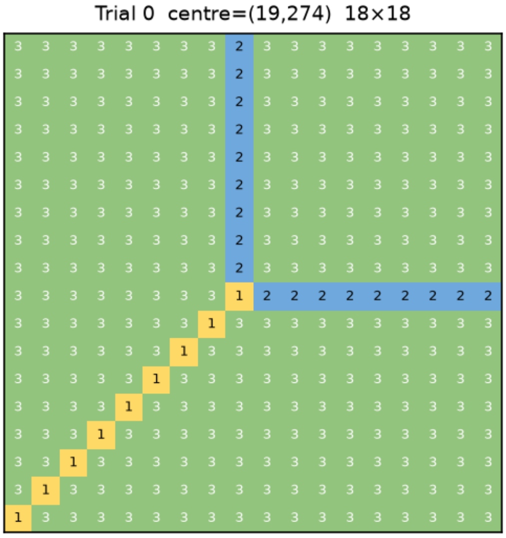

Figure 3.2: Intersection of $(1, 0)$ and $(0, 1)$, yielding $(1, 1)$.

The intersection of $(1, 1)$ and $(0, 1)$ yields $(1, 2)$.

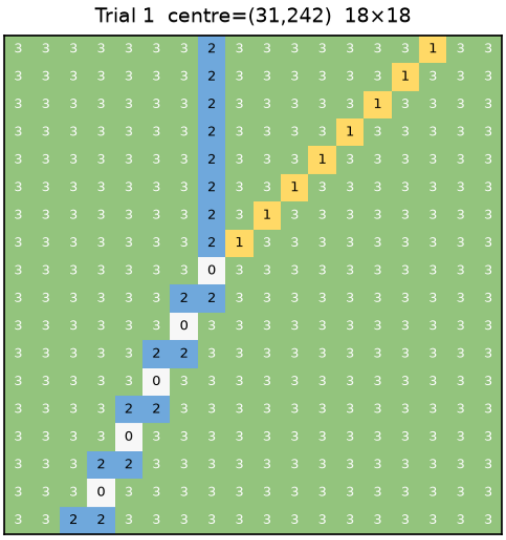

Figure 3.3: Intersection of $(1, 1)$ and $(0, 1)$, yielding $(1, 2)$.

The intersection of $(1, 2)$ and $(0, 1)$ yields $(1, 3)$.

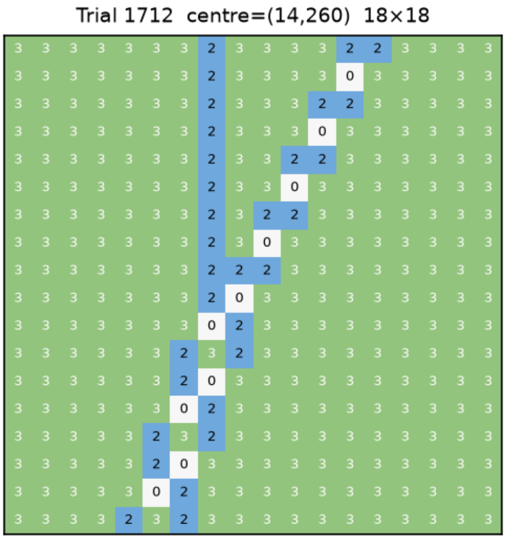

Figure 3.4: Intersection of $(1, 2)$ and $(0, 1)$, yielding $(1, 3)$.

Four lines may also meet at one point. For example, $(0, 1)$, $(2, 1)$, $(1, 1)$, and $(1, 1)$ meet at a point, since $(0, 1) + (2, 1) = (1, 1) + (1, 1)$.

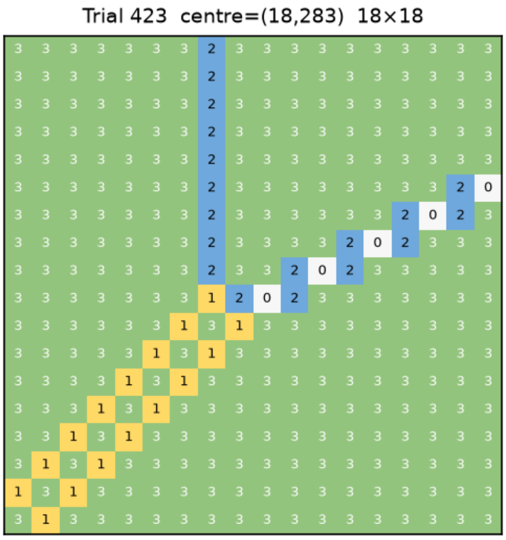

Figure 3.5: Four-line intersection: $(0, 1)$, $(2, 1)$, $(1, 1)$, $(1, 1)$. The direction vectors balance pairwise.

3.3.1 Explaining the Observed Patterns

The zero-sum rule at intersections explains several earlier observations.

No irrational slopes. Every directional line is born at an intersection of two existing lines, so its direction vector is an integer sum (or difference) of the two parent vectors. Starting from $(1, 0)$ and $(0, 1)$, repeated addition can only produce integer vectors (n, m). Hence every slope m/n is rational. An irrational slope would require an irrational vector component, which integer arithmetic can never produce. This explains why no line with an irrational slope appeared in any of the 10 000 trials.

Rarity of near-horizontal or near-vertical lines. A line with slope $1/n$ (or n) for large n requires building a direction vector with very different components. Under the zero-sum rule, this means adding the basic vertical direction repeatedly to a nearly horizontal line, which demands several successive intersections along the same path. Each extra intersection is rare, so directions such as $(1, 3)$ or $(3, 1)$ appear far less often than $(1, 1)$, which results from a single intersection of the two most common lines, $(1, 0)$ and $(0, 1)$. By contrast, the horizontal, vertical, and 45° directions arise from the simplest sums and appear almost everywhere.

3.4 Squared-Norm Rule

For each directional line, let $\mathcal{D}$ be the total height difference from the background over one full period, and let $v = (n, m)$ with $\gcd(n, m) = 1$ be its direction vector. Across all trials and all observed directions,

$$\mathcal{D} = |v|^2 = n^2 + m^2. \tag{3.1}$$

We verify this for four representative directions.

For $v = (0, 1)$, the repeating unit is a single cell of height 2, so $\mathcal{D} = 3 - 2 = 1 = 0^2 + 1^2$.

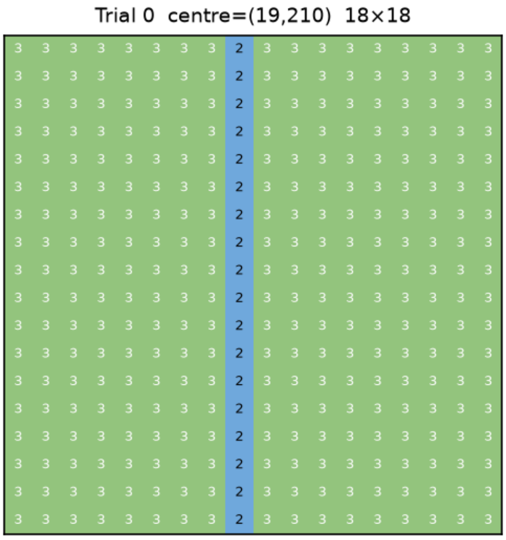

Figure 3.6: Directional pattern $(0, 1)$. Repeating unit: one cell of height 2, $\mathcal{D} = 1$.

For $v = (1, 1)$, the repeating unit is a single cell of height 1, so $\mathcal{D} = 3 - 1 = 2 = 1^2 + 1^2$.

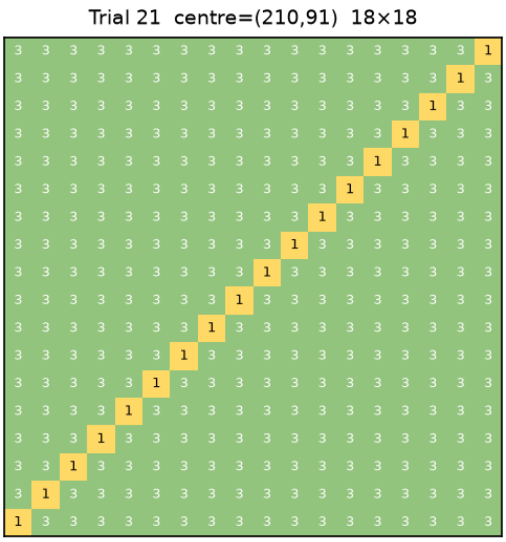

Figure 3.7: Directional pattern $(1, 1)$. Repeating unit: one cell of height 1, $\mathcal{D} = 2$.

For $v = (2, 1)$, the repeating unit consists of three cells of heights $2, 0, 2$, so

$$\mathcal{D} = (3 - 2) + (3 - 0) + (3 - 2) = 5 = 2^2 + 1^2.$$

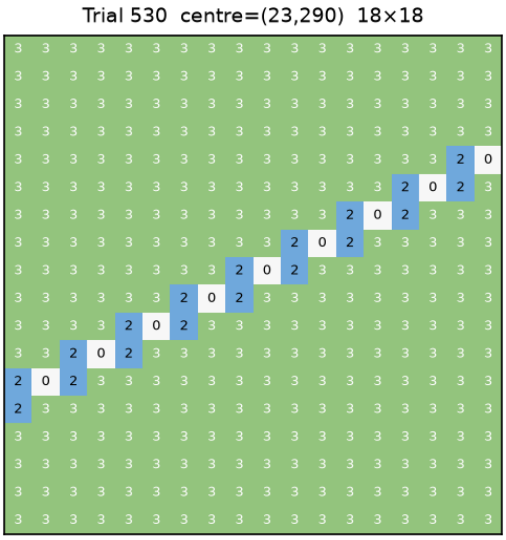

Figure 3.8: Directional pattern $(2, 1)$. Repeating unit: heights $2, 0, 2$, $\mathcal{D} = 5$.

For $v = (3, 1)$, the repeating unit consists of seven cells of heights $2, 2, 0, 3, 0, 2, 2$, so

$$\mathcal{D} = (3 - 2) + (3 - 2) + (3 - 0) + (3 - 3) + (3 - 0) + (3 - 2) + (3 - 2) = 10 = 3^2 + 1^2.$$

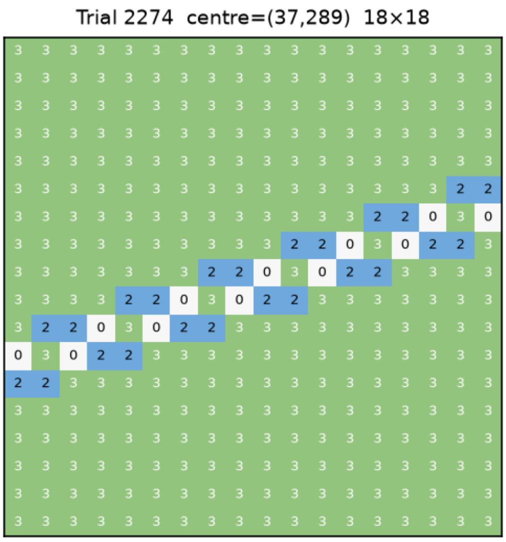

Figure 3.9: Directional pattern $(3, 1)$. Repeating unit: heights $2, 2, 0, 3, 0, 2, 2$, $\mathcal{D} = 10$.

3.4.1 Trivial Case

Each background cell can be viewed as a directional line with vector $(0, 0)$, its repeating unit being the cell itself. The difference from the background is $\mathcal{D} = 3 - 3 = 0 = 0^2 + 0^2$, which also satisfies the rule.